

Econ 501:

Practice question (12-17)

Amjad Sin

A utility f^n is continuous if arbitrarily small changes in x , results in only small change in $U(x)$, no sudden jumps or discontinuity.

Q12: Proof that if a preference relation satisfies monotonicity and continuity, then there exist a utility function $U(\cdot)$ representing such preference relation.

Ans:

If a preference relation satisfies monotonicity and continuity, then there exists a utility function $U(\cdot)$ representing such preference relation.

Proof

Take a bundle $x \neq 0$.

① By monotonicity, $x \succeq 0$, where $0 = (0, 0, \dots, 0)$.
That is, if bundle $x \neq 0$, it contains positive amounts of at least one good and it is preferred to bundle 0.

② Let $m \equiv \max \{x_1, x_2, \dots, x_n\}$ be the number of units of the most abundant good in bundle (x) .
if $x = (2 \text{ apple}, 5 \text{ banana}, 3 \text{ carrot})$ then $m \equiv \max \{2, 5, 3\} = 5$
 m is the amount of the good I have most in my basket.

Define bundle M as the bundle where all components coincide with the highest component of bundle x . That is, $M = (m, m, \dots, m)$ Highest amount of good
if $x = (5 \text{ op}, 5 \text{ ba}, 5 \text{ ca})$ then $M = (5 \text{ op}, 5 \text{ ba}, 5 \text{ ca})$

Hence, by monotonicity, $M \succeq x$.

③ Bundle 0 and M are both on the main diagonal, since each of them contains the same amount of good (x_1) and (x_2) .
 $0(0, 0)$ equal amount of x_1, x_2
 $M(m, m)$ equal amount of x_1, x_2 .

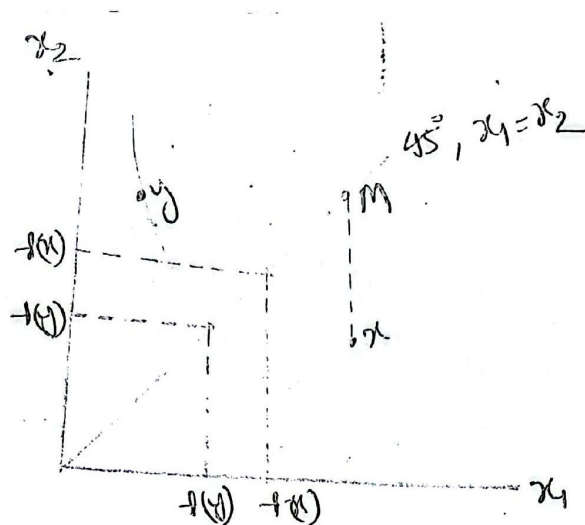


Fig: Existence of a utility function

By continuity and monotonicity, there exists a bundle that is indifferent to x and which lies on the main diagonal. By monotonicity, this bundle is unique. Otherwise, modifying any of its components would lead to higher/lower indifference curves.

Denote such bundle as $(f(x), f(x), \dots, f(x))$.

Let $u(x) = f(x)$, which is a real number.

Applying the same steps to another bundle ($y \neq x$) (see this bundle in the fig), we obtain,
 $(f(y), f(y), \dots, f(y))$.

and let $u(y) = f(y)$, which is also a real number.

Start

13, 14, 15

we know that,

$$x \sim (t(x), t(x), \dots, t(x))$$

$$y \sim (t(y), t(y), \dots, t(y))$$

$$x \succeq y$$

Hence, by transitivity, $x \succeq y$ if

$$x \sim (t(x), t(x), \dots, t(x)) \succeq (t(y), t(y), \dots, t(y)) \sim y$$

and by monotonicity;

$$x \succeq y \Leftrightarrow t(x) \geq t(y) \Leftrightarrow u(x) \geq u(y)$$

Note: A utility function can satisfy continuity but still be non-differentiable.

For instance, the Leontief function utility function, $A \min(ax_1, bx_2)$, is continuous but cannot be differentiated at the kink..

- o -

ch-1 complete.

ch 2 → Demand Theory

ch 2 Demand Theory

[UMP = utility maximization problem]

Q13: Consider a utility maximization problem. If the utility function is continuous and preferences satisfy LNS over the consumption set $X = \mathbb{R}_+^N$, then examine the properties satisfied by the Walrasian demand function $x(p, w)$. Examine the situation when the sufficient condition of UMP may be violated.

Ans:

Consumer maximizes his utility level by selecting a bundle x , subject to his budget constraint:

$$\begin{aligned} \max_{x \geq 0} u(x) \\ \text{s.t. } p \cdot x \leq w \end{aligned}$$

If the utility function is continuous and preferences satisfy LNS over the consumption set $X = \mathbb{R}_+^N$, then the

Walrasian demand $x(p, w)$ satisfies: (properties of Walrasian demand)

① Homogeneity of degree zero:

$$x(p, w) = x(\alpha p, \alpha w) \text{ for all } p, w \text{ and for all } \alpha > 0.$$

That is, the budget set is unchanged.

$$\{x \in \mathbb{R}_+^N : p \cdot x \leq w\} = \{x \in \mathbb{R}_+^N : \alpha p \cdot x \leq \alpha w\}.$$

We don't need any assumption on the preference relation

LNS: local non-satiation

to show this. We only rely on the budget set being unaffected.

Note that, the preference relation can be linear and homothetic would still hold.

$$\frac{q_1 w}{q_2 p_2} = \frac{w}{p_2}$$

$x(p, w)$ is unaffected

[In the fig. budget line $B_{p, w}$, with intercepts

$$\frac{q_1 w}{c_1 p_1} = \frac{w}{p_1}$$

w/p_1 and w/p_2 and illustrates how a common increase in price and wealth produces budget line $B_{p', q', w}$ with exactly same intercepts $\frac{q_1 w}{q_1 p_1} = \frac{w}{p_1}$ as budget line $B_{p, w}$. $B_{p, w}$ and $B_{p', q', w}$ exactly coincide.]

③ Walras' Law:

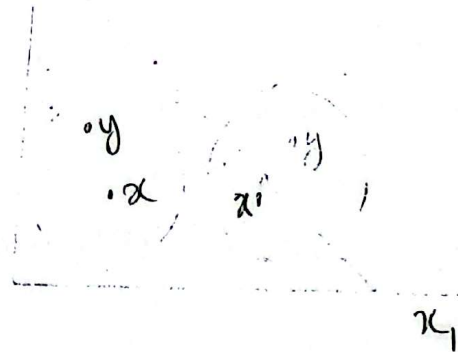
$$p \cdot x = w \text{ for all } x = x(p, w)$$

It follows from LNS: if the consumer selects a Walrasian demand $x \in x(p, w)$, where $p \cdot x < w$, then it means we can still find other bundle y , which is ϵ -close to x , where consumer can improve his utility level.

If the bundle the consumer chooses lies on the budget line, i.e., $p \cdot x' = w$, we could then identify bundles that are strictly preferred to x' , but these bundles would be unaffordable to the consumer.

For $x \in x(p, w)$, there is a bundle y , ϵ -close to x , such that $y \succ x$. Then $x \notin x(p, w)$

π_2



অথবা Walras's Law - নিম্নের অংশে বর্ণনা করা যাবে, $p \cdot x = w$

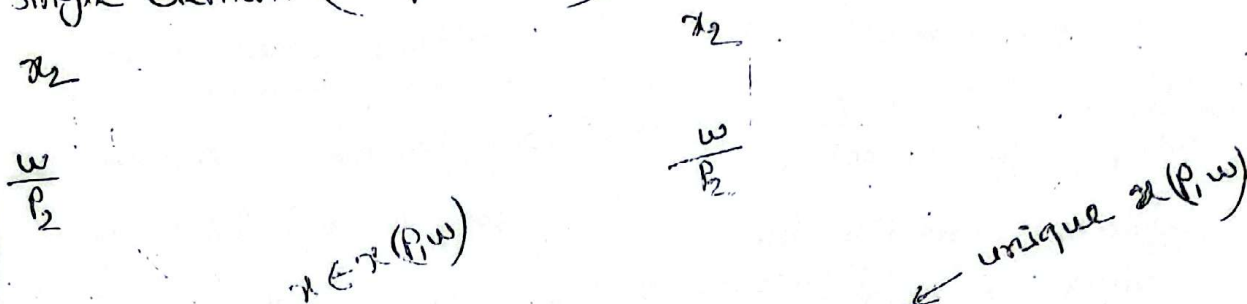
Walras's law follows from LNS. Indeed, if the consumer were to select a bundle that lies strictly inside the budget set (so that he is not exhausting his wealth, point x), we could find another point y at ϵ -distance from bundle x that is strictly preferred by the individual to bundle x . In this case, the initial bundle x would not be utility maximizing because there are other bundles that are still affordable on the budget line x' and we could ^{still identify bundles that} are strictly preferred to x' , ^{northeast of the budget line(y)}. However, these bundles would be unaffordable to the consumer and thus can't be utility maximizing. In summary, utility maximizing bundles, such as x' , must lie on the budget line, thus exhausting all wealth.

③ Convexity / Uniqueness:

① If the preferences are convex, then the Walrasian demand correspondence $x(p, w)$ defines a convex set, i.e. a continuum of bundles are utility maximizing. The budget line and indifference curve can become tangent along several bundles forming a convex set.

1) If the preferences are strictly convex, then:

Walrasian demand correspondence $x(p, w)$ contains a single element (unique bundle): unique tangency point between budget line and indifference curve.



(a)

Fig: Convex preference

(b)

Fig: strictly convex preference.

UMP: Violation of Sufficient Condition

① The consumer

① $u(\cdot)$ is ^{not} monotone:

The consumer chooses bundle A (at a corner) since it yields the highest utility level given his budget constraint.

At point A, however, the tangency condition,

$MRS_{1,2} = \frac{p_1}{p_2}$ does not hold since

consumer would like to further decrease his consumption of x_1 simultaneously. implies that if we increase both goods, we reach a higher utility level, which means that if we increase the amount of some goods, the consumer is not made worse off.

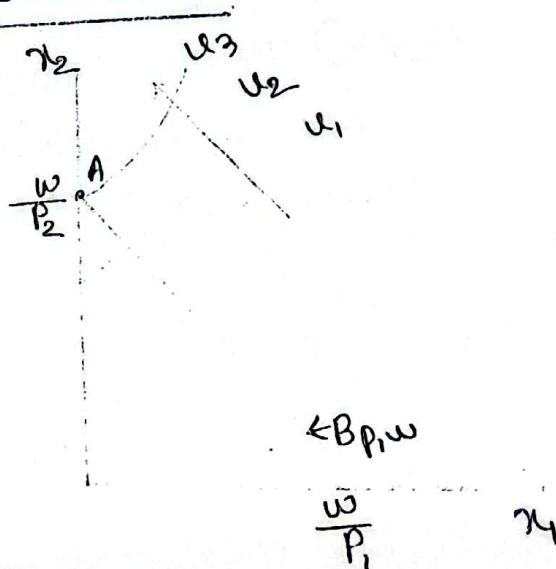
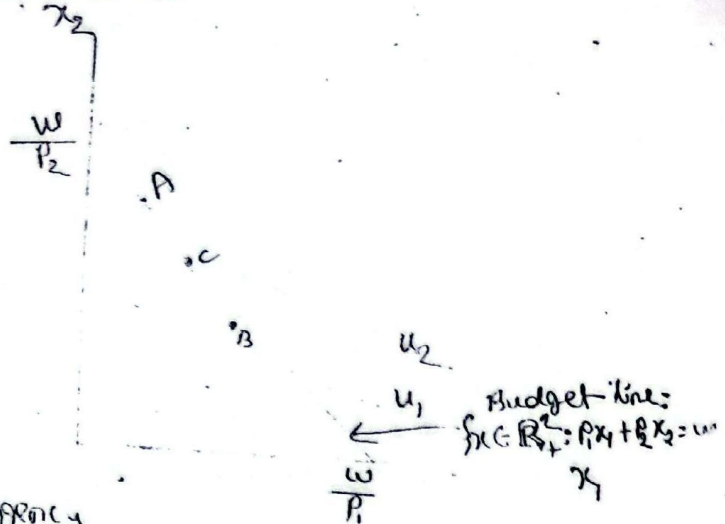


Fig: corner solution

(c) is not quasiconcave:

The upper contour set of c is not convex.
 $MRS_{1,2} = \frac{p_1}{p_2}$ is not a sufficient condition for a max. A point of tangency (c) gives a lower utility level than a point of non-tangency.

(B). The true maximum is at point A. Because the three requirements for necessary conditions to become sufficient conditions are relatively mild (utility function being quasi-concave, monotonic and $\nabla u \neq 0$).



Q.14) Define indirect utility function. Examine its properties.

Ans:

Indirect Utility Function

The Walrasian demand function, $x(p, w)$, is the solution to the UMP (i.e. $\arg\max$).

When we insert the optimal consumption bundle into the individual's utility function, we obtain the highest utility level that the individual can achieve by solving his UMP, $v(x(p, w))$, or compactly referred to as the indirect utility function and denoted by $v(p, w)$.

Or,

~~what would~~ Indirect utility function (i.e. the highest utility level), $v(p, w) \in \mathbb{R}$, associated with the UMP, would be the utility function evaluated at the solution of the UMP, i.e. $x(p, w)$. It is the "value function" of this optimization problem.

Properties of Indirect Utility Function

If the utility function is continuous and preferences satisfy LNS over the consumption set $X = \mathbb{R}_+^L$, then the indirect utility function $v(p, w)$ satisfies:

① Homogeneous of degree zero:

Increasing p and w by a common factor $\alpha > 0$, does not modify the consumer's optimal consumption bundle, $x(p, w)$, nor his maximal utility level, measured by $v(p, w)$.

or,

This property states that increasing market prices and wealth by the same proportion does not modify the consumer's budget set (see fig).

As a consequence, such an increase does not modify the consumer's optimal consumption bundle $x(p, w)$, nor does it affect the maximal utility level that the individual reaches in his UMP, as measured by $v(p, w)$, meaning $v(\alpha p, \alpha w) = v(p, w)$.

$$\frac{\alpha w}{\alpha p} = \frac{w}{p}$$

$$\leftarrow v(p, w) = v(\alpha p, \alpha w)$$

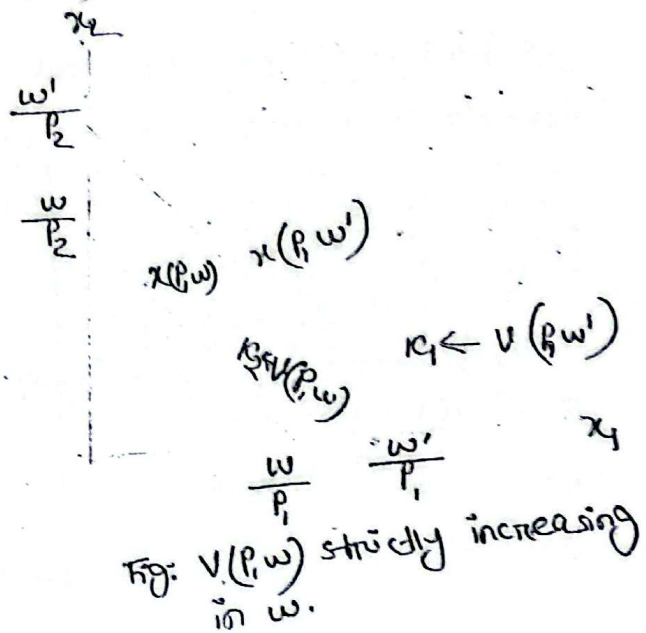
$$\leftarrow x(p, w) = x(\alpha p, \alpha w)$$

$$\leftarrow B_{p, w} = B_{\alpha p, \alpha w}$$

fig: $v(p, w)$ homogeneous of degree zero.

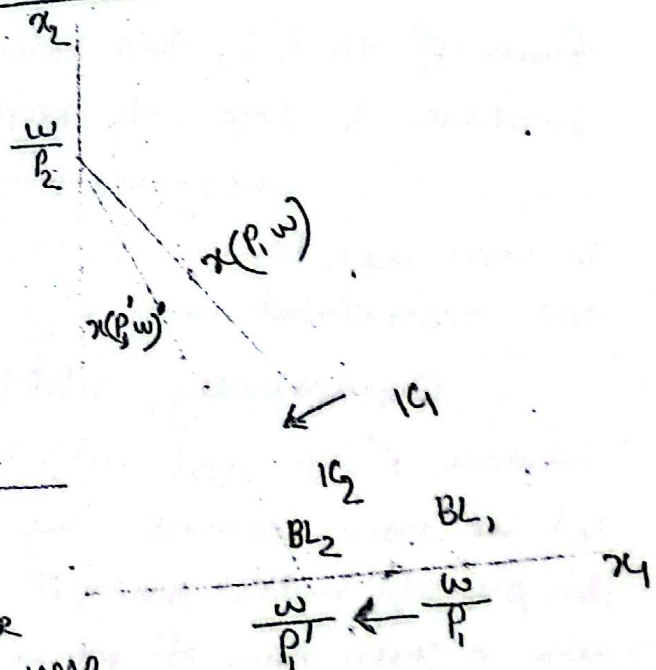
② Strictly increasing in w : $v(p, w') > v(p, w)$ for $w' > w$

This property states that if we increase the wealth of the individual, we enlarge the set of feasible bundles he can afford and, as a consequence, we allow him to reach an indifference curve associated with a higher utility level. Therefore, the maximal utility level that he can obtain is strictly increasing in his wealth, as depicted in the fig. where $v(p, w') > v(p, w)$ because $w' > w$.



③ Non-decreasing (i.e. weakly decreasing) in p_k :

With an increase in the price of any good, the set of affordable bundles shrinks, so the individual can only reach indifference curves associated with weakly lower utility levels. Hence, an increase in the price of any good k produces a weak reduction in the maximal utility level that the individual can obtain by solving his UMP.

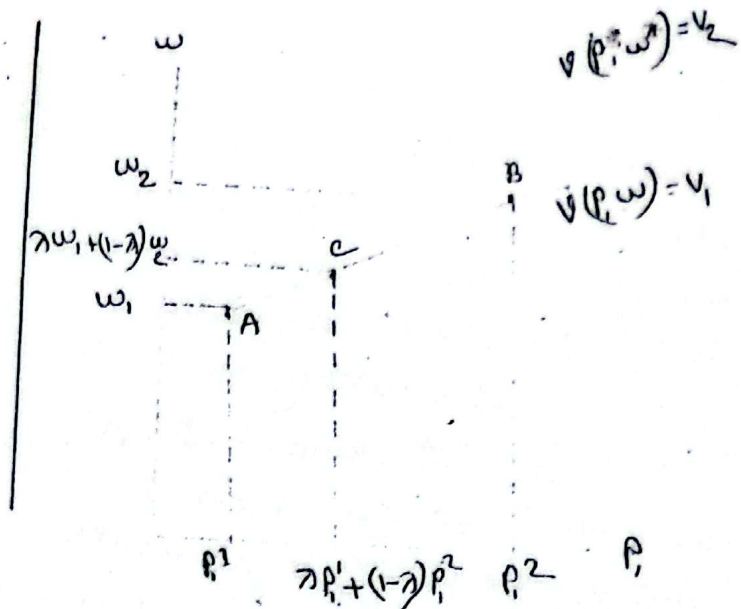


① Quasi-convexity: The set $\{(p, w) : v(p, w) \leq \bar{v}\}$ is convex for any $\bar{v}$.

Interpretation - I:

If $(p^1, w^1) \succeq^* (p^2, w^2)$,
then $(p^1, w^1) \succeq^* (\lambda p^1 + (1-\lambda)p^2, \lambda w^1 + (1-\lambda)w^2)$;
i.e. if $A \succeq^* B$, then

$A \succeq^* C$.



or, The set $\{(p, w) : v(p, w) \leq \bar{v}\}$ is convex for any $\bar{v} \in \mathbb{R}_+$. The figure depicts the indirect utility function with the price of good 1 on the horizontal axis and wealth level on the vertical axis. When price increases from p^1 to p^2 , this individual's also wealth must also increase in order to keep his utility level unaffected. Graphically utility level $v(p^*, w^*) > v(p, w)$ where, in contrast, lower prices and higher wealth levels are associated with higher maximal utilities.

Graphically, utility level $v(p^*, w^*) > v(p, w)$

where $p^* < p$ and $w^* > w$.

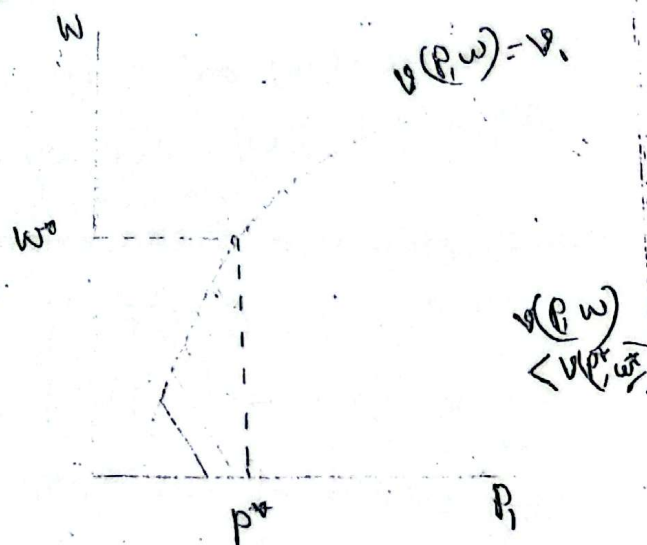
Let us now examine the quasi-convexity of $v(p, w)$. This property implies that, if the maximum utility associated with a given pair of prices and wealth (point A) is

weakly higher than the maximum utility associated with another pair of prices and wealth (point B), then the convex combination of points A and B (point C) yields a maximum utility that is weakly lower than that associated with A. That is, if the consumer faces an intermediate price level $\lambda p_1^1 + (1-\lambda) p_1^2$ and an intermediate wealth level $\lambda w_1 + (1-\lambda) w_2$, the maximum utility level that he will be able to reach by solving his UMP is lower than that at point A $\equiv (p_1^1, w_1)$.

Alternative Interpretation of quasi-convexity

Interpretation II:

$v(p, w)$ is quasiconvex if the set of (p, w) pairs for which $v(p, w) < v(p^*, w^*)$ is convex.

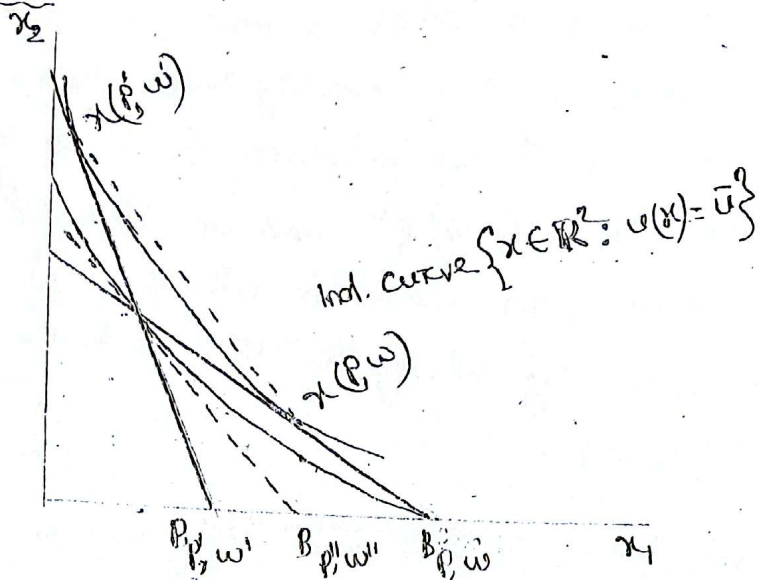


Another interpretation of Quasi-convexity

Interpretation III:

Using x_1 and x_2 in the axis, perform following steps:

- ① When $B_{p, w}$ then $x(p, w)$ bundle
- ② When $B_{p', w'}$ then $x(p', w')$



- ③ Both $x(p, w)$ and $x(p', w')$ induce an indirect utility of $v(p, w) = v(p', w') = \bar{u}$.

- ④ Construct a linear combination of prices and wealth:

$$\left. \begin{aligned} p'' &= \alpha p + (1-\alpha)p' \\ w'' &= \alpha w + (1-\alpha)w' \end{aligned} \right\} B_{p'', w''}$$

- ⑤ Any solution to the VMP given $B_{p'', w''}$ must lie on a lower indifference curve (i.e. lower utility).

$$v(p'', w'') \leq \bar{u}.$$

Q-15) Consider the Cobb-Douglas utility function:

$$U(x_1, x_2) = k x_1^\alpha x_2^{1-\alpha}$$

for some $\alpha \in (0, 1)$ and $k > 0$.

i) Derive the Walrasian demand functions and examine their properties.

ii) How do you get the indirect utility function. Explain their properties. Verify Roy's identity.

iii) Solve the expenditure ~~maximization~~ ^{minimization} problem and explain the properties of the compensated demand function.

Answers:

i) Deriving Walrasian Demand Function

$$U(x_1, x_2) = k x_1^\alpha x_2^{1-\alpha}$$

$$\text{Budget constraint: } p_1 x_1 + p_2 x_2 = W$$

$$\begin{array}{l} W = \text{wealth} \\ p = \text{price} \\ \text{Walrasian demand} \\ \max. U(x) \\ \text{s.t. } px \leq W \end{array}$$

Setting up the Lagrangian,

$$L = k x_1^\alpha x_2^{1-\alpha} + \lambda [W - p_1 x_1 - p_2 x_2]$$

FONC requires $\frac{\partial L}{\partial x_1} = k \alpha x_1^{\alpha-1} x_2^{1-\alpha} - \lambda p_1 = 0 \dots \textcircled{I}$

$$\frac{\partial L}{\partial x_1} = k \alpha x_1^{\alpha-1} x_2^{1-\alpha} - \lambda p_1 = 0 \dots \textcircled{I}$$

$$\frac{\partial L}{\partial x_2} = k x_1^\alpha (1-\alpha) x_2^{-\alpha} - \lambda p_2 = 0 \dots \textcircled{II}$$

$$\frac{SL}{SN} = W - P_{x_1} x_1 - P_{x_2} x_2 = 0 \dots (iii)$$

Dividing Eq. (i) by Eq. (ii),

$$\frac{K \alpha x_1^{\alpha-1} x_2^{1-\alpha}}{K x_1^\alpha (1-\alpha) x_2^{-\alpha}} = \frac{x P_{x_1}}{x P_{x_2}}$$

$$\Rightarrow \frac{\alpha}{1-\alpha} x_1^{-1} x_2^{1-\alpha+\alpha} = \frac{P_{x_1}}{P_{x_2}}$$

$$\Rightarrow \frac{\alpha}{1-\alpha} x_1^{-1} x_2 = \frac{P_{x_1}}{P_{x_2}}$$

$$\Rightarrow \frac{\alpha}{1-\alpha} \cdot \frac{x_2}{x_1} = \frac{P_{x_1}}{P_{x_2}}$$

$$\Rightarrow P_{x_2} x_2 = P_{x_1} x_1 \cdot \frac{1-\alpha}{\alpha} \dots (iv)$$

Substituting Eq. (iv) into Eq. (iii),

$$W - P_{x_1} x_1 - P_{x_2} x_2 = 0$$

$$\Rightarrow W - P_{x_1} x_1 - P_{x_1} x_1 \cdot \frac{1-\alpha}{\alpha} = 0$$

$$\Rightarrow W - P_{x_1} x_1 \left(1 + \frac{1-\alpha}{\alpha}\right) = 0$$

$$\Rightarrow W - P_{x_1} x_1 \left(\frac{\alpha+1-\alpha}{\alpha}\right) = 0$$

$$\Rightarrow W - P_{x_1} \cdot x_1 \cdot \frac{1}{\alpha} = 0$$

$$\Rightarrow W = P_{x_1} \cdot x_1 \cdot \frac{1}{\alpha}$$

$$\Rightarrow P_{x_1} \cdot x_1 = \alpha W$$

$$\Rightarrow x_1 = \frac{\alpha W}{P_{x_1}}$$

$$\text{Similarly, } x_2 = \frac{(1-\alpha)W}{P_{x_2}}$$

So, Walrasian demand for goods $x_1 = \frac{\alpha W}{P_{x_1}}$ and $x_2 = \frac{(1-\alpha)W}{P_{x_2}}$
 The demand for every good k is increasing in wealth w , decreasing in its own price, P_k , but unaffected by the price of other good, P_l , where $l \neq k$.

Properties of Walrasian demand

- ① Homogeneity of degree zero $x(p, w) = x(\alpha p, \alpha w)$ for all $p \gg 0$, $w > 0$ and for all $\alpha > 0$.
- ② Walras's law, $p \cdot x(p, w) = w$
- ③ Convexity/uniqueness: If preferences are convex, then the Walrasian demand $x(p, w)$ defines a convex set. If preferences are strictly convex, then the Walrasian demand $x(p, w)$ contains a single element.

$\rightarrow$ Details $\Rightarrow$ Q-13 to 17

11.11

Homogeneity of degree: ~~zero~~

$$\begin{aligned} U(x_1, x_2) &= K x_1^\alpha x_2^{1-\alpha} \\ &= K (cx_1)^\alpha (cx_2)^{1-\alpha} \\ &= K \cdot c^\alpha x_1^\alpha c^{1-\alpha} x_2^{1-\alpha} \\ &= c^{\alpha+1-\alpha} \cdot K x_1^\alpha x_2^{1-\alpha} \\ &= c K x_1^\alpha x_2^{1-\alpha} \end{aligned}$$

$$\left. \begin{aligned} x_1(p, w) &= x_1 \\ x_2(p, w) &= x_2 \end{aligned} \right\} \begin{array}{l} \text{(1) - ମୂଳ} \\ \text{ସମସ୍ୟା} \\ \text{ନିର୍ଦ୍ଦିଷ୍ଟ ରୁ} \end{array}$$

(i) Indirect utility function: $v(p, w)$

Walrasian demands for good 1 and 2 are,

$$x_1(p, w) = \frac{\alpha w}{p_1}$$

$$x_2(p, w) = \frac{(1-\alpha)w}{p_2}$$

Indirect utility function:

$$\begin{aligned} v(p, w) &= K(x_1)^\alpha \cdot x_2^{1-\alpha} \\ &= K\left(\frac{\alpha w}{p_1}\right)^\alpha \cdot \left(\frac{(1-\alpha)w}{p_2}\right)^{1-\alpha} \\ &= K\left(\frac{\alpha}{p_1}\right)^\alpha \cdot w^\alpha \cdot \left(\frac{1-\alpha}{p_2}\right)^{1-\alpha} \cdot w^{1-\alpha} \\ &= K\left(\frac{\alpha}{p_1}\right)^\alpha \cdot \left(\frac{1-\alpha}{p_2}\right)^{1-\alpha} \cdot w^{\alpha+1-\alpha} \\ &= K\left(\frac{\alpha}{p_1}\right)^\alpha \cdot \left(\frac{1-\alpha}{p_2}\right)^{1-\alpha} \cdot w \end{aligned}$$

This indirect utility function is decreasing in the price of both goods, increasing in the consumer's wealth, w and homogeneous of degree zero, since increasing all prices and wealth by a common factor A yields,

$$\begin{aligned} v(Ap, Aw) &= K\left(\frac{\alpha}{Ap_1}\right)^\alpha \cdot \left(\frac{1-\alpha}{Ap_2}\right)^{1-\alpha} (Aw) \\ &= K \cdot \left(\frac{\alpha}{p_1}\right)^\alpha \cdot A^{-\alpha} \cdot \left(\frac{1-\alpha}{p_2}\right)^{1-\alpha} \cdot A^{1-\alpha} \cdot A \cdot w \\ &= A^{-\alpha+1-\alpha+1} \cdot K\left(\frac{\alpha}{p_1}\right)^\alpha \cdot \left(\frac{1-\alpha}{p_2}\right)^{1-\alpha} \cdot w \\ &= A^0 \cdot v = v \quad [A^0 = 1] \end{aligned}$$

Homogeneity of degree 0.

Properties of indirect utility function $\Rightarrow$ see (9-14)

Roy's Identity:

Relationship between the Walrasian demand and the indirect utility function:

Consider a continuous utility function representing a strictly convex preference relation that satisfies LNS. Additionally suppose that the indirect utility function associated with the VMP problem is differentiable at any pair of prices and wealth. We then obtain

Roy's Identity:

$$-\frac{\frac{Sv(p, w)}{Sp_k}}{\frac{Sv(p, w)}{Sw}} = x_k(p, w) \text{ for every good } k.$$

In our example,

$$\begin{aligned} v &= k \left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{1-\alpha}{p_2} \right)^{1-\alpha} w \\ \frac{Sv(p, w)}{Sp_1} &= \left(\alpha \cdot \frac{1}{p_1} \right)^\alpha \cdot k \left(\frac{1-\alpha}{p_2} \right)^{1-\alpha} w \\ &= \alpha \cdot \alpha \cdot \left(\frac{1}{p_1} \right)^{\alpha-1} \cdot k \left(\frac{1-\alpha}{p_2} \right)^{1-\alpha} w \\ &= \alpha \cdot \left(\frac{1}{p_1} \right)^{\alpha-1} \cdot \frac{d}{dp_1} \left(\frac{1}{p_1} \right) \cdot k \left(\frac{1-\alpha}{p_2} \right)^{1-\alpha} w \\ &= \alpha \left(\frac{1}{p_1} \right)^{\alpha-1} \cdot \ln \frac{1}{p_1} \cdot k \left(\frac{1-\alpha}{p_2} \right)^{1-\alpha} w \end{aligned}$$

In our example, (K = constant)

$$V = K \left(\frac{\alpha}{P_1} \right)^{\alpha} \left(\frac{1-\alpha}{P_2} \right)^{1-\alpha} W$$

$$= K \cdot \alpha^{\alpha} P_1^{-\alpha} (1-\alpha)^{1-\alpha} P_2^{1-\alpha} W$$

$$\eta_1 = - \frac{\frac{SV}{SP_1}}{\frac{SV}{SW}}$$

$$= - \frac{(-\alpha) P_1^{-\alpha-1} \cdot K \cdot \alpha^{\alpha} (1-\alpha)^{1-\alpha} P_2^{1-\alpha} W}{P_1^{-\alpha} \cdot K \cdot \alpha^{\alpha} (1-\alpha)^{1-\alpha} P_2^{1-\alpha} W}$$

$$= -(-\alpha) P_1^{-\alpha-1+1} W$$

$$= \frac{\alpha W}{P_1}$$

Same as,

$$\eta_2 = - \frac{\frac{SV}{P_2}}{\frac{SV}{SW}} = \frac{(1-\alpha) W}{P_2}$$

(ii) Given that,

$$u(x_1, x_2) = kx_1^\alpha x_2^{1-\alpha}$$

Solve Expenditure Minimization Problem:

pr
u > u

Consider a consumer with Cobb-Douglas utility function, $u(x_1, x_2) = kx_1^\alpha x_2^{1-\alpha}$. We can find the Hicksian demands for good 1 and good 2 by solving the following expenditure minimization problem (EMP):

$$\min_{x_1, x_2 \geq 0} p_1 x_1 + p_2 x_2 \quad \text{subject to } kx_1^\alpha x_2^{1-\alpha} \geq u$$

FOMC requires,

Lagrangian Function:

$$L = p_1 x_1 + p_2 x_2 + \lambda [u - kx_1^\alpha x_2^{1-\alpha}]$$

Taking first order necessary condition (FOMC):

$$\frac{\partial L}{\partial x_1} = p_1 - \alpha (x_1^{\alpha-1}) \lambda k x_2^{1-\alpha} = 0$$

$$\Rightarrow p_1 = \lambda k \alpha x_1^{\alpha-1} x_2^{1-\alpha} \quad \text{--- (1)}$$

$$\frac{\partial L}{\partial x_2} = p_2 - \lambda k (1-\alpha) x_1^\alpha x_2^{-\alpha} = 0$$

$$\Rightarrow p_2 = \lambda k (1-\alpha) x_1^\alpha x_2^{-\alpha} \quad \text{--- (2)}$$

$$\frac{\partial L}{\partial \lambda} = u - kx_1^\alpha x_2^{1-\alpha} = 0$$

$$\Rightarrow kx_1^\alpha x_2^{1-\alpha} = u \quad \text{--- (3)}$$

where λ denotes the Lagrange multiplier of the minimization problem. In the case of interior solutions, the first order conditions above become

$$P_1 = \lambda K \alpha x_1^{\alpha-1} x_2^{1-\alpha} \dots \quad (4)$$

$$P_2 = \lambda K (1-\alpha) x_1^\alpha x_2^{-\alpha} \dots \quad (5)$$

$$K x_1^\alpha x_2^{1-\alpha} = u \quad \dots \quad (6)$$

Divide Eq. (4) by Eq. (5),

$$\frac{P_1}{P_2} = \frac{\lambda K \alpha x_1^{\alpha-1} x_2^{1-\alpha}}{\lambda K (1-\alpha) x_1^\alpha x_2^{-\alpha}}$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{\alpha}{1-\alpha} \cdot x_1^{-1+\alpha} x_2^{1-\alpha+\alpha}$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{\alpha}{1-\alpha} \cdot \frac{x_2}{x_1}$$

$$\Rightarrow \frac{(1-\alpha) P_1 x_2}{\alpha P_2} = x_1 \quad \dots \quad (7)$$

Substituting Eq. (7) into Eq. (6),

$$K x_1^\alpha \left[\frac{(1-\alpha) P_1 x_2}{\alpha P_2} \right]^{1-\alpha} = u$$

$$\Rightarrow K x_1^\alpha x_2^{1-\alpha} \left(\frac{(1-\alpha) P_1}{\alpha P_2} \right)^{1-\alpha} = u$$

$$\Rightarrow x_1^{\alpha+1-\alpha} = \frac{u}{K \left[\frac{(1-\alpha) P_1}{\alpha P_2} \right]^{1-\alpha}} = u \times \frac{1}{K} \times \frac{(\alpha P_2)^{1-\alpha}}{((1-\alpha) P_1)^{1-\alpha}}$$

$$\Rightarrow x_1 = \left[\frac{u \frac{1}{K} \frac{\alpha P_2}{(1-\alpha) P_1}}{1} \right]^{1-\alpha}$$

Similarly, $x_2 = \left(\frac{(1-\alpha)p_1}{\alpha p_2} \right)^\alpha \cdot u \cdot \frac{1}{k}$

Thus, simultaneously solving for x_1 and x_2 , we find that the Hicksian demands for goods 1 and 2 are,

$h_1(p, u) = \frac{1}{k} \left(\frac{\alpha p_2}{(1-\alpha)p_1} \right)^{1-\alpha} \cdot u$ and

$h_2(p, u) = \frac{1}{k} \left(\frac{(1-\alpha)p_1}{\alpha p_2} \right)^\alpha \cdot u$

Therefore the expenditure function is

$e(p, u) = p_1 \cdot h_1(p, u) + p_2 \cdot h_2(p, u)$

$$\begin{aligned} &= p_1 \cdot \frac{1}{k} \left(\frac{\alpha p_2}{(1-\alpha)p_1} \right)^{1-\alpha} \cdot u + p_2 \cdot \frac{1}{k} \left(\frac{(1-\alpha)p_1}{\alpha p_2} \right)^\alpha \cdot u \\ &= \frac{1}{k} \left(\frac{\alpha}{1-\alpha} \right)^{1-\alpha} p_1^{1-\alpha} p_2^{1-\alpha} \cdot u + \frac{1}{k} \left(\frac{1-\alpha}{\alpha} \right)^\alpha p_1^\alpha p_2^\alpha \cdot u \\ &= \frac{1}{k} p_1^\alpha p_2^{1-\alpha} \cdot u \left[\left(\frac{\alpha}{1-\alpha} \right)^{1-\alpha} + \left(\frac{1-\alpha}{\alpha} \right)^\alpha \right] \end{aligned}$$

which is homogeneous of degree one in prices, strictly increasing in u . In addition, $e(p, u)$ is strictly increasing in the prices of every good k , p_k and at a decreasing rate, thus satisfying concavity in prices.

② $\frac{1}{k} \cdot p_1^\alpha p_2^{1-\alpha} u$

Ch-3 (Demand Theory Application)

Q16: (a) Distinguish between CV and EV.

(b) How do you measure the welfare effects of a price change?

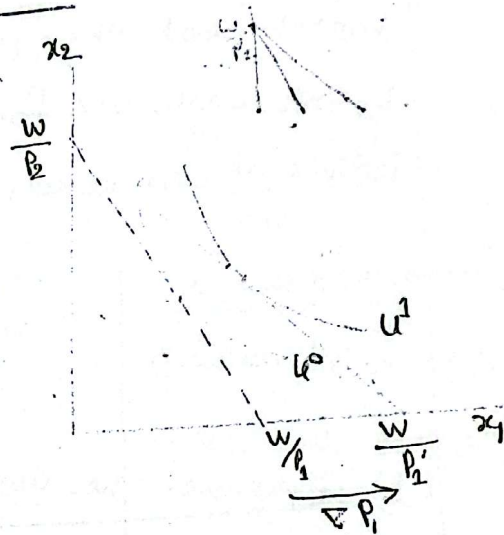
Ans: Measure the welfare effects of a price change - इस में
वर्तमान स्थिति
का कारण
निर्धार

(a) Distinguishing between CV and EV

Compensating Variation (CV):

This welfare measures how much
money would a consumer be
willing to give up after a reduction
in prices to be just as well off
as before the price decrease.
 For this reason we refer this
 measure as the "compensating
 variation," since it represents the
 change in income that would
 exactly compensate the consumer
 for the impact that the price
 change has on his welfare.

Importantly, this measure maintains
 the initial utility level, U^0 (before
 any price change).



The fig. represents a decrease
 in the price of x_1 , from P_1 to P_1'
 yielding an increase in the utility
 level the consumer reaches
 after solving his utility
 maximization problem (UMP),
 from U^0 to U^1 .

→ (ii) compensating CV as fig. 11
 → figure 11 shows CV and EV
 page-9 CV as fig. 11
 → Fig. price decrease and utility
 relation change

② Equivalent variation (EV): It responds to the question, How much money would a consumer need before a reduction in prices to be as well off as after the price decrease? For this reason we refer to this measure as the "equivalent variation," since it reflects the change in income that would be equivalent to the impact that the price change has on his welfare. In this case, we focus on the final utility level that the individual can reach after price change, u^1 .

(b) Measure the welfare effects of a price change

Two approaches:

① CV and EV using expenditure function

② CV and EV using Hicksian demand

① CV using Expenditure Function:

$CV(p^0, p^1, w)$ using $e(p, u)$, expenditure function we get,

$$CV(p^0, p^1, w) = e(p^1, u^1) - e(p^1, u^0)$$

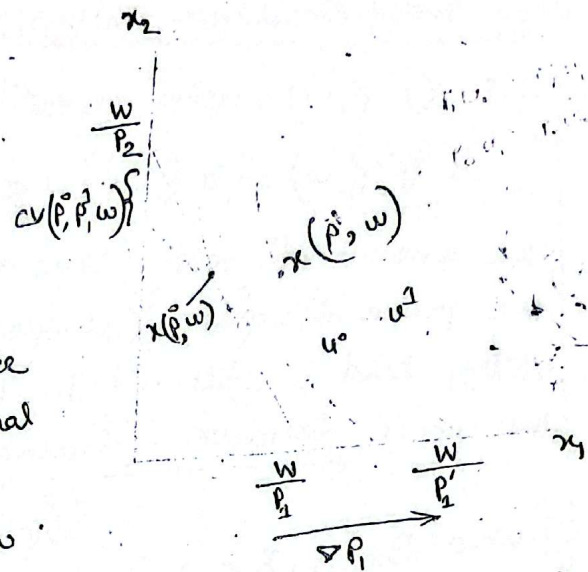
The amount of money the consumer is willing to give up after the price decrease (after price level is p^1 and

here utility level has improved to u^1 to be just as well off as before the price decrease (reaching utility level u^0).

The figure represents the compensating variation using the expenditure function.

Here is a basic procedure on how to find the CV from a price decrease. Before the price change, the consumer selects optimal bundle $x(p^0, w)$ reaching utility level u^0 under budget set $B_{p^0, w}$.

After p^1 decreases, the consumer chooses optimal bundle $x(p^1, w)$ under budget set $B_{p^1, w}$. We then adjust the consumer's wealth to make him just as well off as before the price change. We represent this adjustment by a parallel shift in budget line $B_{p^1, w}$, to the dashed budget line, where the individual can reach his initial utility level u^0 at the final price ratio. At this point his expenditure is $e(p^1, u^0)$. Hence the difference in expenditure $e(p^1, u^1)$ at $B_{p^1, w}$ and $e(p^1, u^0)$ at the dashed line represents the compensating variation resulting from the price change.



In short (simpler),

- ① when $B_{p^0, w}$ & $x(p^0, w)$
 - ② Δp and $x(p^1, w)$ under $B_{p^1, w}$
 - ③ Adjust final wealth (after the price change) to make the consumer as well off as before the price change.
 - ④ Difference in expenditure:

$$CV(p^0, p^1, w) = e(p^1, u^1) - e(p^1, u^0)$$

at $B_{p^1, w}$ dashed line
- This is Hicksian wealth compensation.

EV using Expenditure Function:

EV (p^0, p^1, w) , using expenditure function $e(p, u)$:

$$EV(p^0, p^1, w) = e(p^0, u^1) - e(p^0, u^0).$$

The amount of money the consumer needs to receive before the price decrease (at the initial price level p^0 when her utility level is still u^0) to be just as well off as after the price decrease (reaching utility level u^1).

① when $B(p^0, w)$, $x(p^0, w)$

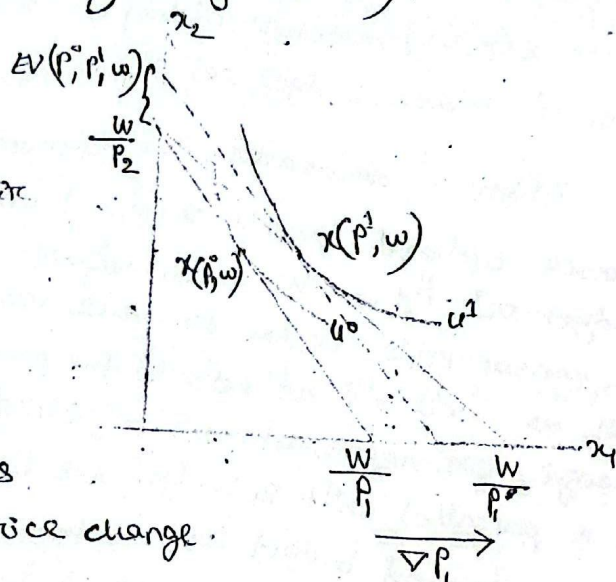
② ∇p_1 and $x(p^1, w)$ under $B(p^1, w)$.

③ Adjust initial wealth (before the price change)

to make the consumer as well off as after the price change.

④ Difference in expenditure:

$$EV(p^0, p^1, w) = \underbrace{e(p^0, u^1)}_{\text{dashed line}} - \underbrace{e(p^0, u^0)}_{\text{at } B(p^0, w)}$$



③ CV using Hicksian Demand :

From the previous definitions we know that if $p_1^1 < p_1^0$ and $p_k^1 = p_k^0$ for all $k \neq 1$, then

$$\begin{aligned} CV(p_1^0, p_1^1, w) &= e(p_1^1, u^0) - e(p_1^0, u^0) \\ &= w - e(p_1^1, u^0) \\ &= e(p_1^0, u^0) - e(p_1^1, u^0) \\ &= \int_{p_1^1}^{p_1^0} \frac{\partial e(p_1, \bar{p}_{-1}, u^0)}{\partial p_1} dp_1 \\ &= \int_{p_1^1}^{p_1^0} h_1(p_1, \bar{p}_{-1}, u^0) dp_1 \end{aligned}$$

$$e(p_1^1, u^0) = w$$

Since $e(p_1^1, u^1) = e(p_1^0, u^0) = w$ and $\frac{\partial e(p_1, u)}{\partial p_1} = h_1(p_1, u)$.

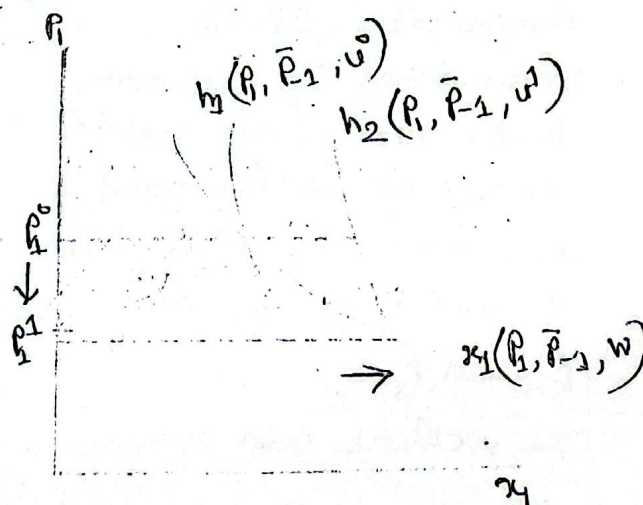
In addition, let $\bar{p}_{-1}$ denote the price vector of all other goods $k \neq 1$ such that $\bar{p}_{-1} = (p_2, p_3, \dots, p_n)$, which are unaffected by the price change.

The case is :

- Normal good
- Price decrease

Graphically, CV is represented by the area to the left of the Hicksian demand curve for good 1 associated with utility level u^0 and lying between prices p_1^1 and p_1^0 .

The welfare gain is represented by the shaded region.



EV using Hicksian Demand:

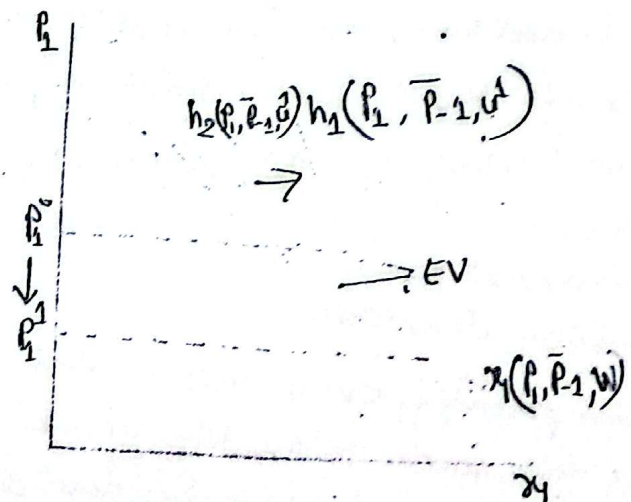
From the previous definitions we know that, if $p_1^1 < p_1^0$ and $p_k^1 = p_k^0$ for all $k \neq 1$, then

$$\begin{aligned} EV(p^0, p^1, w) &= e(p^0, u^1) - e(p^1, u^1) \\ &= e(p^0, u^1) - w \\ &= e(p^0, u^1) - e(p^1, u^1) \\ &= \int_{p_1^1}^{p_1^0} \frac{se(p_1, \bar{p}_{-1}, u^1)}{sp_1} dp_1 \\ &= \int_{p_1^1}^{p_1^0} h_1(p_1, \bar{p}_{-1}, u^1) dp_1 \end{aligned}$$

The case is:

- Normal good
- Price decrease

Graphically, EV is represented by the area to the left of the Hicksian demand curve for good 1 associated with utility level u^1 and lying between prices p_1^1 and p_1^0 .



The welfare gain is represented by the shaded region.

Normally price increases
change at

What about a price increase?

The Hicksian demand associated with initial utility level u^0 (before the price increase or before the introduction of a tax) experiences an inward shift when the price increases, or consumer's utility level is now u^1 , where $u^0 > u^1$. Hence,

$$h_1(p_1, \bar{p}_{-1}, u^0) > h_1(p_1, \bar{p}_{-1}, u^1)$$

The definitions of CV and EV would now be:

CV: the amount of money that a consumer would need after a price increase to be as well off as before the price increase.

EV: the amount of money that a consumer would be willing to give up before a price increase to be as well off as after the price increase.

Q17: Graphically show that for a decrease in the price of a commodity (which both goods are normal).

$$EV > CS > CV$$

for price increase,

$$EV < CS < CV$$

Ans:

Waltasian demand is easier to observe, so we could use the variation in consumer's surplus as an approximation of welfare changes. A decrease in p_1 would imply an increase in CS . This measurement coincides with that provided by the CV and EV , when income effects are zero.

Recall that the Waltasian demand measures both income and substitution effects resulting from a price change,

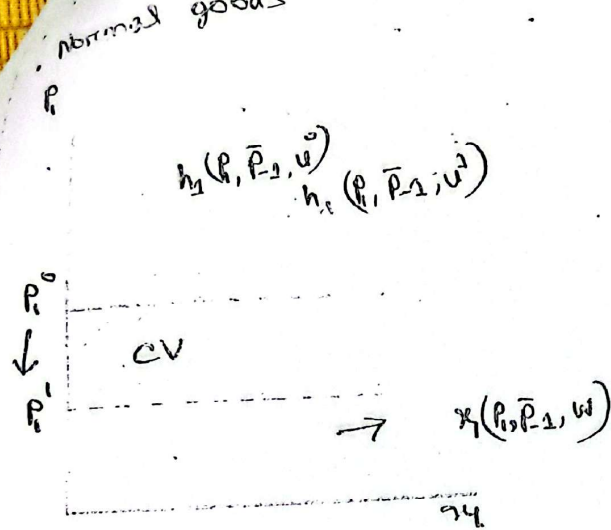
while,

the Hicksian demand measures only substitution effects from such a price change.

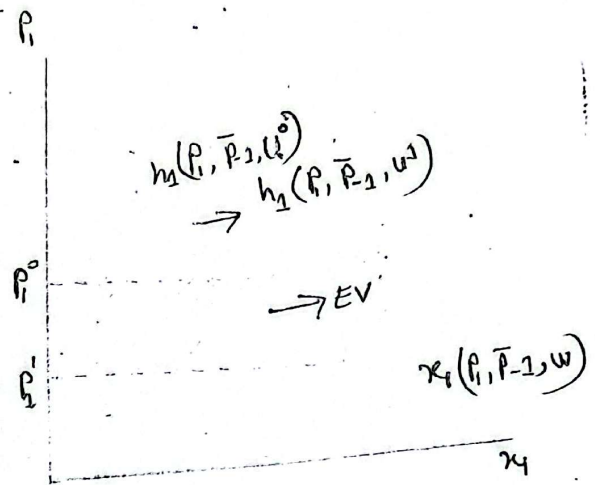
Hence, there will be a difference between CV and CS , and between EV and CS .

The figure illustrates the difference between these welfare measures for the case in which good 1 is a "normal" good, namely income effects are positive. Hence, the Waltasian demand is more sensitive to price changes than the Hicksian demand, implying that $CV < CS$ whereas $EV > CS$, meaning $EV > CS > CV$.

normal goods



(a) $CV < CS$



(b) $CS < EV$

For normal goods:

Price decrease: $CV < CS < EV$.

Price increase: $CV > CS > EV$.

Consumer Surplus, $CS = \int_{P_1}^{P_0} x_1(P, \bar{P}_{-1}, m) dP$

For inferior good

Price decrease: $EV < CS < CV$

Price increase: $EV > CS > CV$